Elizabeth "Eli" Margolin

ecmargo@seas.upenn.edu · ecmargo.github.io · github.com/ecmargo · Google Scholar

SUMMARY

PhD candidate at the University of Pennsylvania building zero-knowledge proof systems at the intersection of programming languages and cryptography, with publications at IEEE S&P, USENIX Security, and SOSP. Three years of security engineering at Meta. Current work applies these tools to AI accountability, including zkTLS provenance for LLM responses.

TECHNICAL SKILLS

Cryptography: zero-knowledge proofs; secure two-party and multi-party computation; homomorphic encryption; differential privacy.
Programming: Rust, C++, Python; interactive theorem proving (Rocq, Lean).
Security: threat modeling; abuse and insider-threat detection; security-sensitive code review.

EXPERIENCE

University of Pennsylvania — PhD Researcher, Cryptography & Programming Languages *2021–present*

- Architected and implemented zero-knowledge proof systems for formal languages and zkTLS protocols that let users prove facts about private data and real web sessions
- Designed and built a differentially private federated analytics system that leverages secure multi-party computation and homomorphic encryption to scale to over a billion devices
- Work published at IEEE S&P, USENIX Security, and SOSP
- Advised by Sebastian Angel. Dissertation: *Zero Knowledge Proofs of Formal Languages and Their Applications*

Brave Software — Research Intern *Summer 2024*

- Designed zero-knowledge protocols for privacy-preserving fraud detection over authenticated web traffic, taking research designs through to deployable implementations

Meta Platforms — Security Engineer *2019–2021 (Contingent Worker, 2022)*

- Performed threat modeling and security analysis to design and own company-wide insider-threat and abuse-detection tooling for security and legal investigations teams
- Scaled alert-processing pipelines through automation and led security and privacy reviews with Legal and Policy, raising throughput while cutting false positives and data footprint

SELECTED PUBLICATIONS

Coral: Fast Succinct Non-Interactive Zero-Knowledge CFG Proofs *IEEE S&P 2026*
S. Angel, S. Celi, E. Margolin*, P. Mishra, M. Sander, J. Woods

Reef: Fast Succinct Non-Interactive Zero-Knowledge Regex Proofs *USENIX Security 2024*
S. Angel, E. Ioannidis, E. Margolin*, S. Setty, J. Woods

Arboretum: A Planner for Large-Scale Federated Analytics with Differential Privacy *SOSP 2023*
E. Margolin, K. Newatia, E. Roth, T. Luo, A. Haeberlen

Weasel: Zero Knowledge Proofs of WASM Compilation *In submission*
J. Woods, E. Margolin, E. Ioannidis, S. Angel, P. Mishra

Surf: Bringing zkTLS to the Modern Web *In submission*
S. Angel, S. Celi, E. Margolin*

** Authors listed in alphabetical order.*

EDUCATION

University of Pennsylvania — PhD, Computer and Information Science *In progress*

Duke University — MS, Economics and Computation

Stanford University — BA, Political Science, with Honors in International Security

SELECTED ACTIVITIES

Team USA — National Team Athlete (Rowing); USRowing Athlete Council *2025–present*

- 1st place, 2026 World Rowing Cup III, Lucerne
- A Finalist, 2025 World Rowing Championships, Shanghai